A MONIC POLYNOMIAL OVER A RING IS DIVISIBLE BY T MINUS EACH OF ITS ROOTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Division with remainder by a monic polynomial over an arbitrary ring. used at proof lines 4561-4562, where P is written as (T - alpha)Q plus a remainder r in R. Produces the decomposition whose remainder has to be shown to vanish.
- Evaluation at alpha is a ring homomorphism R[T] to R and is additive. used at proof lines 4562-4564 when P(alpha) and (alpha - alpha)Q(alpha) + r are compared. Turns the hypothesis P(alpha) = 0 into the vanishing of the remainder r.
- A monic polynomial of degree one is T minus its root. used at proof lines 4559-4560 for the base case of the induction. Settles the case deg P = 1 and starts the induction.

Source reference. Original lemma 052Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6587–6587. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. Let $P(T)$ be a monic polynomial with coefficients in R. Let $\alpha \in R$ be such that $P(\alpha) = 0$. Then $P(T) = (T - \alpha)Q(T)$ for some monic polynomial $Q(T) \in R[T]$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By induction on the degree of P. If $\deg(P) = 1$, then $P(T) = T - \alpha$ and the result is true. If $\deg(P) > 1$, then we can write $P(T) = (T - \alpha)Q(T) + r$ for some polynomial $Q \in R[T]$ of degree $< \deg(P)$ and some $r \in R$ by long division. By assumption $0 = P(\alpha) = (\alpha - \alpha)Q(\alpha) + r = r$ and we conclude that $r = 0$ as desired. $\square$